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  "creator" : "ari.neko@cvut.cz",
  "timestamp" : "05-07-2026_13:36:13",
  "variant" : 1,
  "subjectName" : "mathematics",
  "categories" : [ "Stereometry", "Planimetry", "Probability" ],
  "questions" : [ {
    "question" : "What is the volume of a cone with base radius r and height h?",
    "answers" : [ "2 * pi * r * h", "(1/3) * pi * r^2 * h", "(4/3) * pi * r^2 * h", "pi * r^2 * h" ],
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    "question" : "What is the volume formula for a perfect sphere with radius r?",
    "answers" : [ "2 * pi * r", "4 * pi * r^2", "(4/3) * pi * r^3", "pi * r^2" ],
    "correctIndex" : 2
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    "question" : "What is the sum of the interior angles of a regular hexagon?",
    "answers" : [ "900 degrees", "360 degrees", "540 degrees", "720 degrees" ],
    "correctIndex" : 3
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    "question" : "What is the area of a circle with a diameter of 10?",
    "answers" : [ "10 * pi", "25 * pi", "50 * pi", "100 * pi" ],
    "correctIndex" : 1
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    "question" : "According to Brahmagupta's formula, what does it calculate?",
    "answers" : [ "Area of a cyclic quadrilateral", "Volume of a pyramid", "Area of a scalene triangle",
    "Perimeter of an ellipse" ],
    "correctIndex" : 0
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    "question" : "Which theorem states that the square of the hypotenuse is equal to the sum of
    squares of the other two sides?",
    "answers" : [ "Euclid's Theorem", "Thales's Theorem", "Pythagorean Theorem", "Ptolemy's
    Theorem" ],
    "correctIndex" : 2
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    "question" : "What is the probability of flipping a fair coin three times and getting all heads?",
    "answers" : [ "1/6", "1/4", "1/8", "1/2" ],
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    "question" : "What is the probability of rolling a sum of 7 with two fair six-sided dice?",
    "answers" : [ "1/4", "1/12", "1/36", "1/6" ],
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"question" : "What is the sum of probabilities of all possible mutually exclusive outcomes in a sample space?",

"answers" : ["1", "Infinite", "100", "0"],

"correctIndex" : 0

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"question" : "If events A and B are independent, what is $P(A \text{ and } B)$?",

"answers" : [" $P(A) + P(B) - P(A \text{ or } B)$ ", " $P(A) / P(B)$ ", " $P(A) + P(B)$ ", " $P(A) * P(B)$ "],

"correctIndex" : 3

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}

